

## Homogeneous Linear Systems

1. A system of linear equations is said to be **homogeneous** if the constant terms are all zero; that is, the system has the form

$$\begin{aligned}x + 4y + 5z &= 0 \\ 2x + 5y - 10z &= 0 \\ 3x + 4y - 51z &= 0\end{aligned}$$

It's obvious that  $(x, y, z) = (0, 0, 0)$  is one solution of the system

*Trivial Solution*

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0$$

...

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0$$

$\therefore$  There must be at least one soln.  
So, the system must be consistent

2. Every homogeneous system of linear equations is **consistent** because all such systems have  $x_1 = 0, x_2 = 0, \dots, x_n = 0$  as a solution. This solution is called the **trivial solution**; if there are other solutions, they are called **nontrivial solutions**.

$$(x, y, z) \neq (0, 0, 0)$$

3. Because a homogeneous linear system always has the trivial solution, there are **only two possibilities** for its solutions:

$$(x, y, z) = (0, 0, 0)$$

(a) The system has **only the trivial solution**. *[unique soln]*

(b) The system has **infinitely many solutions** in addition to the trivial solution. *[Infinite]*

4. **Free Variable Theorem for Homogeneous Systems:**

If a homogeneous linear system has  $n$  unknowns, and if the reduced row echelon form (RREF) of its augmented matrix has  $r$  nonzero rows, then the system has  $(n - r)$  free variables.

[Note: A free variable is a variable that can be treated as a parameter.]

## Equivalent Statements

If  $A$  is an  $n \times n$  matrix, then the following statements are equivalent.

- (a)  $A$  is invertible.
- (b)  $\det(A) \neq 0$ .
- (c)  $Ax = 0$  has only the **trivial solution**.
- (d) The reduced row echelon form (RREF) of  $A$  is  $I_n$ .
- (e)  $Ax = b$  is **consistent** for every  $n \times 1$  matrix  $b$ .
- (f)  $Ax = b$  has **exactly one solution** for every  $n \times 1$  matrix  $b$ .

$$c) \left( A \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A^{-1}A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\therefore$  Homogeneous

$$e) \left( A \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$A^{-1}A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} \text{fixed} \\ \text{fixed} \\ \text{fixed} \end{pmatrix}$$

$\therefore$  System is consistent

$$f) \left( A \right) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$A^{-1}A \begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} \text{fixed} \\ \text{fixed} \\ \text{fixed} \end{pmatrix}$$

$\therefore$  One solution

**Example 6** (CIE 2020s21/8(a))Find the values of  $a$  for which the system of equations

$$\begin{aligned} 3x + y + z &= 0, \\ ax + 6y - z &= 0, \\ ay - 2z &= 0, \end{aligned}$$

Homogeneous

does not have a unique solution.

[3]

(Ans:  $a = 4$  or  $a = -9$ )**Solution**

For the system to not have a unique solution, c) is false

$$\Rightarrow \text{Det} = 0$$

$$3 \begin{vmatrix} 6 & -1 \\ a & -2 \end{vmatrix} - a \begin{vmatrix} 1 & 1 \\ a & -2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 3 & 1 & 1 \\ a & 6 & -1 \\ 0 & a & -2 \end{vmatrix} = 0$$

$$3(-12+a) - a(-2-a) = 0$$

$$a^2 + 5a - 36 = 0$$

$$(a+9)(a-4) = 0$$

$$a = -9 \text{ or } a = 4$$

**Example 7** (CIE 2014w11/5)Find the value of  $a$  for which the system of equations

$$x - y + 2z = 4,$$

$$x + ay - 3z = b,$$

$$x - y + 7z = 13,$$

where  $a$  and  $b$  are constants, has no unique solution.

[3]

(Ans:  $a = -1$ )Taking  $a$  as the value just found,

- (i) find the general solution in the case
- $b = -5$
- ,

[4]

$$\left( \text{Ans: } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0 \\ 1.8 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right)$$

- (ii) interpret the situation geometrically in the case
- $b \neq -5$
- .

[1]

(Ans: planes form a prism)

**Solution**

For system to have no unique soln, (f) false

$$\Rightarrow \text{Determinant} = 0$$

$$\begin{vmatrix} 1 & -1 & 2 \\ 1 & a & -3 \\ 1 & -1 & 7 \end{vmatrix} = 0$$

$$1 \begin{vmatrix} a & -3 \\ -1 & 7 \end{vmatrix} + 1 \begin{vmatrix} 1 & -3 \\ 1 & 7 \end{vmatrix} + 2 \begin{vmatrix} 1 & a \\ 1 & -1 \end{vmatrix} = 0$$

$$1(7a - 3) + 1(7 + 3) + 2(-1 - a) = 0$$

$$5a + 5 = 0$$

$$\boxed{a = -1} \quad \#$$

## Solution for Example 7 (cont.)

Taking  $a$  as the value just found, *Infinite solutions*(i) find the general solution in the case  $b = -5$ ,

$$\left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 1 & -1 & -3 & -5 \\ 1 & -1 & 7 & 13 \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & -5 & -9 \\ 0 & 0 & 5 & 9 \end{array} \right]$$

$$\text{Ans: } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0.4 \\ 0 \\ 1.8 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad [4]$$

$$\xrightarrow{R_3 \rightarrow R_3 + R_2} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & -5 & -9 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\xrightarrow{R_2 \rightarrow -\frac{1}{5}R_2} \left[ \begin{array}{ccc|c} 1 & -1 & 2 & 4 \\ 0 & 0 & 1 & 9/5 \\ 0 & 0 & 0 & 0 \end{array} \right] \quad \text{REF}$$

We have  $z = 9/5$ 

$$x - y + 2z = 4$$

Let  $y = t$ , where  $t \in \mathbb{R}$ 

$$\text{Then } x - t + 2(9/5) = 4 \Rightarrow x = \frac{2}{5} + t$$

 $\therefore$  General soln is

$$x = \frac{2}{5} + t, y = t, z = \frac{9}{5}, t \in \mathbb{R}$$

(ii) interpret the situation geometrically in the case  $b \neq -5$ .

[1]

(Ans: planes form a prism)

$$\vec{n}_1 = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

 $\therefore$  No 2 planes are parallel

$$\vec{n}_2 = \begin{pmatrix} -1 \\ a \\ -3 \end{pmatrix}$$

 $\therefore$  From i) if  $b \neq -5$ , then will

$$\vec{n}_3 = \begin{pmatrix} -1 \\ 1 \\ 7 \end{pmatrix}$$

be a contradiction. Thus, no solution

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = 4$$

$$\vec{n} \cdot \vec{n} = 4$$

 $\therefore$  2 planes parallel & No solution $\Rightarrow$  Prism

**Example 8** (CIE 2016s13/3)Find the two values of the constant  $k$  for which the equations

$$\begin{aligned} kx + y + z &= 2, \\ x + ky + z &= -1, \\ x + y + kz &= -1, \end{aligned}$$

have no unique solution.

[4]

(Ans:  $k = 1$  or  $k = -2$ )Show that, for one of these values of  $k$ , the equations have no solution, and solve the equations for the other value of  $k$ . [3](Ans:  $k = 1$ : inconsistent;  $k = -2$ :  $(x, y, z) = ([-1 + t], t, t)$ )**Solution**For the system to have no unique solution  
 $\Rightarrow$  (f) false $\therefore$  We set determinant = 0

$$\begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix} = 0$$

$$\Rightarrow k \begin{vmatrix} k & 1 \\ 1 & k \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 1 & k \end{vmatrix} + 1 \begin{vmatrix} 1 & k \\ 1 & 1 \end{vmatrix} = 0$$

$$\Rightarrow k(k^2 - 1) - 1(k - 1) + 1(1 - k) = 0$$

$$\Rightarrow k(k+1)(k-1) - (k-1) - (k-1) = 0$$

$$\Rightarrow (k-1)[k(k+1) - 1 - 1] = 0$$

$$\Rightarrow (k-1)(k^2 + k + 2) = 0$$

$$\Rightarrow (k-1)(k-1)(k+2) = 0$$

$$\Rightarrow k = 1 \text{ or } k = 1 \text{ or } k = -2$$

 $\therefore$  The system has no unique  
soln when  $k = 1$  or  $k = -2$ When  $k = 1$ , we have

$$x + y + z = 2 \quad \text{--- ①}$$

and

$$x + y + z = -1 \quad \text{--- ②}$$

① - ②,  $0 = 3$  which is a contradiction $\therefore$  For  $k = 1$ , the eqn has no soln.When  $k = -2$ ,

$$\left[ \begin{array}{ccc|c} -2 & 1 & 1 & 2 \\ 1 & -2 & 1 & -1 \\ 1 & 1 & -2 & -1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_2} \left[ \begin{array}{ccc|c} 1 & -2 & 1 & -1 \\ -2 & 1 & 1 & 2 \\ 1 & 1 & -2 & -1 \end{array} \right]$$

$$\begin{array}{l} R_2 \rightarrow R_2 + 2R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array} \rightarrow \left[ \begin{array}{ccc|c} 1 & -2 & 1 & -1 \\ 0 & -3 & 3 & 0 \\ 0 & 3 & -3 & 0 \end{array} \right]$$

$$R_3 \rightarrow R_3 + R_2 \rightarrow \left[ \begin{array}{ccc|c} 1 & -2 & 1 & -1 \\ 0 & -3 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$R_2 \rightarrow -\frac{1}{3}R_2 \rightarrow \left[ \begin{array}{ccc|c} 1 & -2 & 1 & -1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Let  $z = t$ , where  $t \in \mathbb{R}$ 

$$y - z = 0 \Rightarrow y = z \Rightarrow \boxed{y = t}$$

$$x - 2y + z = -1 \Rightarrow x - 2y - z = -1 \Rightarrow x = 2t - t - 1 = t - 1$$

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 $\therefore x = t - 1, y = t, z = t$ , where  $t \in \mathbb{R}$



## 1.2: EIGENVALUES AND EIGENVECTORS

### Introduction

#### 1. Definition (Eigenvector and eigenvalue):

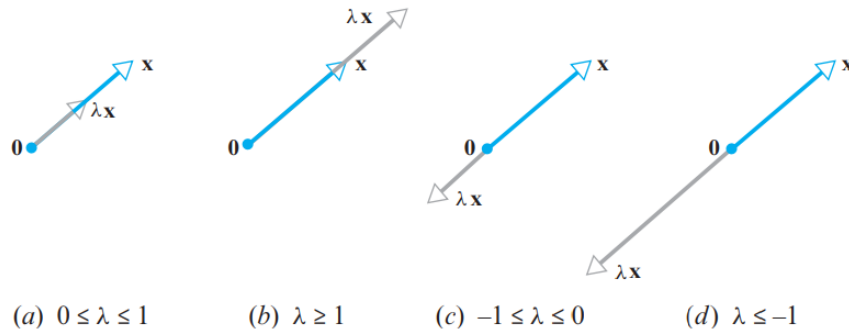
If  $A$  is an  $n \times n$  matrix, then a **nonzero vector**  $\mathbf{x}$  (with order  $n \times 1$ ) is called an **eigenvector** of  $A$  if  $A\mathbf{x}$  is a scalar multiple of  $\mathbf{x}$ ; that is, if

$$A\mathbf{x} = \lambda\mathbf{x}$$

Eigenvector      Eigenvalue

for some scalar  $\lambda$ . The scalar  $\lambda$  is called an **eigenvalue** of  $A$ , and  $\mathbf{x}$  is said to be an eigenvector of  $A$  **corresponding** to  $\lambda$ .

2. In two-dimensional space and three-dimensional space, multiplication by  $A$  maps each eigenvector  $\mathbf{x}$  of  $A$  onto the same line through the origin as  $\mathbf{x}$ . Depending on the sign and the magnitude of the eigenvalue  $\lambda$  corresponding to  $\mathbf{x}$ , the linear operator  $A\mathbf{x} = \lambda\mathbf{x}$  compresses or stretches  $\mathbf{x}$  by a factor of  $\lambda$ , with a reversal of direction in the case where  $\lambda$  is negative.



3. Consider the equation  $A\mathbf{x} = \lambda\mathbf{x}$ .

Now multiply both sides of the equation by the identity matrix to get  $IA\mathbf{x} = I(\lambda\mathbf{x})$ . Then  $A\mathbf{x} = \lambda\mathbf{x}$ , which can be written as

Recall:  
Homogeneous linear system  
$$\begin{pmatrix} \text{coeff.} \\ \text{matrix} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{aligned} A\mathbf{x} &= \lambda\mathbf{x} \\ IA\mathbf{x} &= I(\lambda\mathbf{x}) \\ A\mathbf{x} &= \lambda(I\mathbf{x}) \\ A\mathbf{x} - \lambda I\mathbf{x} &= \mathbf{0} \\ (A - \lambda I)\mathbf{x} &= \mathbf{0} \end{aligned}$$

Homogeneous linear system  
$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$
  
Coeff. matrix Non zero

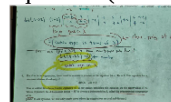
$\therefore$  This is a Homo.L system, but eigen vec. must be non-zero, (cannot be trivial soln). So to have other soln, (c) fail, thus  $\det(A) = 0$

4. For  $\lambda$  to be an eigenvalue, there must be a **nonzero solution** of the equation  $(A - \lambda I)\mathbf{x} = \mathbf{0}$ . This equation has a nonzero solution if and only if

$$A\mathbf{x} \neq \mathbf{0}, \text{ c) false}$$

$$\therefore \det(\text{coeff. matrix}) = 0$$

$$\det(A - \lambda I) = 0$$

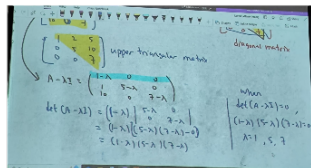


This is called the **characteristic equation of  $A$** ; the scalars satisfying this equation are the **eigenvalues** of  $A$ . When expanded, the determinant  $\det(A - \lambda I)$  is always a polynomial in  $\lambda$ , called the **characteristic polynomial** of  $A$ .

[Note: In our syllabus, we will only study cases where the eigenvalues are real and distinct.]

#### 5. Theorem:

If  $A$  is an  $n \times n$  **triangular matrix** (upper triangular, lower triangular, or diagonal), then the **eigenvalues** of  $A$  are the **entries on the main diagonal** of  $A$ .



6. **Fact:** Any **non-zero scalar multiple** of an **eigenvector** is also an eigenvector **corresponding to the same eigenvalue**.

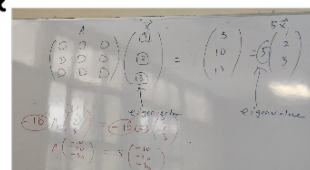
Suppose that  $\lambda$  is an eigenvalue of  $A$  w corresponding eigenvector  $\vec{x}$

$$\text{Then, } A\vec{x} = \lambda\vec{x}$$

$$\Rightarrow k(A\vec{x}) = k(\lambda\vec{x})$$

$$\Rightarrow A(k\vec{x}) = \lambda(k\vec{x})$$

$\therefore k\vec{x}$  is an eigenvector of  $A$  corresponding to eigenvalue  $\lambda$



**Example 10**

Find the eigenvalues and eigenvectors of the matrix  $\mathbf{B} = \begin{pmatrix} -2 & 1 \\ 6 & 3 \end{pmatrix}$ .

[**Note:** Use each eigenvalue to produce a corresponding eigenvector. Remember that any non-zero scalar multiple of an eigenvector is also an eigenvector corresponding to the same eigenvalue.]

$$\left( \text{Ans: } \lambda = -3, \begin{pmatrix} 1 \\ -1 \end{pmatrix}; \lambda = 4, \begin{pmatrix} 1 \\ 6 \end{pmatrix} \right)$$

**Solution**

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} -2-\lambda & 1 \\ 6 & 3-\lambda \end{pmatrix}$$

$$\det(\mathbf{B} - \lambda \mathbf{I}) = 0$$

$$\Rightarrow (-2-\lambda)(3-\lambda) - 6 = 0$$

$$\Rightarrow \lambda^2 - \lambda - 6 - 6 = 0$$

$$\Rightarrow \lambda^2 - \lambda - 12 = 0$$

$$\Rightarrow (\lambda - 4)(\lambda + 3) = 0$$

$$\Rightarrow \boxed{\lambda = 4} \text{ or } \boxed{\lambda = -3}$$

$\therefore$  The eigen values of  $\mathbf{B}$   
are 4 and -3

$$(\mathbf{B} - \lambda \mathbf{I}) \vec{x} = \vec{0}$$

$$\Rightarrow \begin{pmatrix} -2-\lambda & 1 \\ 6 & 3-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \text{ where } \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\Rightarrow \begin{cases} (-2-\lambda)x + y = 0 \\ 6x + (3-\lambda)y = 0 \end{cases}$$

For  $\lambda = 4$

$$\begin{cases} -6x + y = 0 \\ 6x - y = 0 \end{cases}$$

There must be one free variable  
for this system

Let  $x = t$ , then  $y = 6t$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} t \\ 6t \end{pmatrix} = t \begin{pmatrix} 1 \\ 6 \end{pmatrix}$$

But becz we want  
a specific soln

For  $\lambda = 4$ , an eigenvector  
is  $\begin{pmatrix} 1 \\ 6 \end{pmatrix}$

For  $\lambda = -3$

$$\begin{cases} x + y = 0 \\ 6x + 6y = 0 \end{cases}$$

$$x = -y$$

For  $\lambda = -3$ ,

an eigenvector is  $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$

**Example 11**

Find the eigenvalues and corresponding eigenvectors for  $\mathbf{B} = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix}$ .

[Note: Eigenvectors are normally written without fractions.]

Slow Method

Solution

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix}$$

$$\det(\mathbf{B} - \lambda \mathbf{I}) = 0$$

$$\Rightarrow (3-\lambda) \begin{vmatrix} 2-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 0 \\ 1 & 1-\lambda \end{vmatrix} + 4 \begin{vmatrix} 1 & 2-\lambda \\ 1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0 \quad \text{--- ①}$$

$$\text{Let } f(\lambda) = \lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$$

$$f(-1) = -1 - 6 - 5 + 12 = 0$$

$\therefore (\lambda + 1)$  is a factor of  $f(\lambda)$  By inspection

$$\text{So, } f(\lambda) = (\lambda + 1) [\lambda^2 - 7\lambda + 12]$$

$$= (\lambda + 1)(\lambda - 3)(\lambda - 4)$$

Continue from ①,

$$(\lambda + 1)(\lambda - 3)(\lambda - 4) = 0$$

$$\lambda = -1 \text{ or } \lambda = 3 \text{ or } \lambda = 4$$

Exam will always only come out all distinct soln

Butz all are distinct,  
system of solutions will  
have 1 free variable  
[Reason is out of syllabus]

$$(\mathbf{B} - \lambda \mathbf{I}) \vec{x} = \vec{0}$$

$$\Rightarrow \begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \text{ where } \vec{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$$\Rightarrow \begin{cases} (3-\lambda)x + 2y + 4z = 0 \\ x + (2-\lambda)y = 0 \\ x - 2y + (1-\lambda)z = 0 \end{cases}$$

$$\left( \text{Ans: } \lambda = -1, \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}; \lambda = 3, \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}; \lambda = 4, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right)$$

For  $\lambda = -1$  [We know this system has infinite soln]

$$\begin{cases} 4x + 2y + 4z = 0 & \text{--- ①} \\ x + 3y = 0 & \text{--- ②} \\ x - 2y + 2z = 0 & \text{--- ③} \end{cases}$$

From ②,  $x = -3y$  Set this to be free variable

From ③,  $2z = -x + 2y$

$$z = -\frac{1}{2}x + y$$

$$z = -\frac{1}{2}(-3y) + y$$

$$z = \frac{5}{2}y$$

$\therefore$  For  $\lambda = -1$ , an eigenvector is  $\begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$  when  $y = 2$

For  $\lambda = 3$

$$\begin{cases} 2y + 4z = 0 & \text{--- ④} \\ x - y = 0 & \text{--- ⑤} \\ x - 2y - 2z = 0 & \text{--- ⑥} \end{cases}$$

From ⑤,  $x = y$  Set this to be free variable

From ④,  $y + 2z = 0$

$$z = -\frac{1}{2}y$$

$\therefore$  For  $\lambda = 3$ , an eigenvector is  $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$  when  $y = 2$

For  $\lambda = 4$

$$\begin{cases} -x + 2y + 4z = 0 & \text{--- ⑦} \\ x - 2y = 0 & \text{--- ⑧} \\ x - 2y - 3z = 0 & \text{--- ⑨} \end{cases}$$

From ⑧,  $x = 2y$

$\therefore$  For  $\lambda = 4$ , an eigenvector

From ⑦,  $4z = x - 2y$  is  $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

$$\Rightarrow z = \frac{1}{4}x - \frac{1}{2}y$$

$$\Rightarrow z = \frac{1}{4}(2y) - \frac{1}{2}y$$

$$\Rightarrow z = 0$$
 z is fixed, so cannot be chosen to be free

**Example 11**

Find the eigenvalues and corresponding eigenvectors for  $\mathbf{B} = \begin{pmatrix} 3 & 2 & 4 \\ 1 & 2 & 0 \\ 1 & -2 & 1 \end{pmatrix}$ .

[Note: Eigenvectors are normally written without fractions.]

Fast method

$$\left( \text{Ans: } \lambda = -1, \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}; \lambda = 3, \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}; \lambda = 4, \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right)$$

Solution

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} 3-\lambda & 2 & 4 \\ 1 & 2-\lambda & 0 \\ 1 & -2 & 1-\lambda \end{pmatrix}$$

$$\det(\mathbf{B} - \lambda \mathbf{I}) = 0$$

$$\Rightarrow (3-\lambda) \begin{vmatrix} 2-\lambda & 0 \\ -2 & 1-\lambda \end{vmatrix} - 2 \begin{vmatrix} 1 & 0 \\ 1 & 1-\lambda \end{vmatrix} + 4 \begin{vmatrix} 1 & 2-\lambda \\ 1 & -2 \end{vmatrix} = 0$$

$$\Rightarrow \lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0 \quad \text{--- ①}$$

$$\text{Let } f(\lambda) = \lambda^3 - 6\lambda^2 + 5\lambda + 12 = 0$$

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$$= (\lambda + 1)(\lambda - 3)(\lambda - 4)$$

Continue from ①,

$$(\lambda + 1)(\lambda - 3)(\lambda - 4) = 0$$

$$\lambda = -1 \text{ or } \lambda = 3 \text{ or } \lambda = 4$$

Exam will always only come out all distinct soln

Butz all are distinct,  
system of solutions will  
have 1 free variable  
[Reason is out of syllabus]

For  $\lambda = -1$

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} 4 & 2 & 4 \\ 1 & 3 & 0 \\ 1 & -2 & 2 \end{pmatrix}$$

Treat these rows as column vectors

$$\begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}, \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}$$

Take the cross product  
of any 2 non-parallel  
vectors

$$\begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 - 12 \\ -(0 - 4) \\ 12 - 2 \end{pmatrix} \\ = \begin{pmatrix} -12 \\ 4 \\ 10 \end{pmatrix} \\ = 2 \begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$$

$\therefore$  For  $\lambda = -1$ , an eigenvector is  $\begin{pmatrix} -6 \\ 2 \\ 5 \end{pmatrix}$

For  $\lambda = 3$ :

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} 0 & 2 & 4 \\ 1 & -1 & 0 \\ 1 & -2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 + 4 \\ -(0 - 4) \\ 0 - 2 \end{pmatrix} \\ = \begin{pmatrix} 4 \\ 4 \\ -2 \end{pmatrix} \\ = 2 \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$$

$\therefore$  For  $\lambda = 3$ , an eigenvector is  $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$

For  $\lambda = 4$ :

$$\mathbf{B} - \lambda \mathbf{I} = \begin{pmatrix} -1 & 2 & 4 \\ 1 & -2 & 0 \\ 1 & -2 & -3 \end{pmatrix}$$

$$\begin{pmatrix} -1 \\ 2 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ -2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 + 8 \\ -(0 - 4) \\ 2 - 2 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 0 \end{pmatrix} = 4 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$\therefore$  For  $\lambda = 4$ , an eigenvector is  $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$

**Example 12**

Find the eigenvalues and corresponding eigenvectors for  $\mathbf{C} = \begin{pmatrix} 2 & 1 & 1 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$ .

$$\left( \text{Ans: } \lambda = -1, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}; \lambda = 2, \begin{pmatrix} 3 \\ -1 \\ 1 \end{pmatrix}; \lambda = 3, \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix} \right)$$

**Solution**

**Fact:** If a matrix,  $\mathbf{A}$ , has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , then  $\mathbf{A}\mathbf{e} = \lambda\mathbf{e}$ .

**Example 13** (Finding eigenvalue, given the eigenvector)

Given that  $\mathbf{A} = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{pmatrix}$ , show that the following vectors are eigenvectors, and determine their eigenvalues:

$$\mathbf{e}_1 = \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix}, \mathbf{e}_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{e}_3 = \begin{pmatrix} 1 \\ -6 \\ 3 \end{pmatrix}$$

(Ans:  $\lambda_1 = -2, \lambda_2 = 5, \lambda_3 = 1$ )

**Solution**

$$\mathbf{A}\vec{e}_1 = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 4 & -12 \\ 0 \\ 12 & -6 \end{pmatrix} = \begin{pmatrix} -8 \\ 0 \\ 6 \end{pmatrix} = -2 \begin{pmatrix} 4 \\ 0 \\ -3 \end{pmatrix} = -2\vec{e}_1$$

$\therefore \vec{e}_1$  is an eigenvector w corresponding eigenvalue -2

$$\mathbf{A}\vec{e}_2 = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \dots = 5\vec{e}_2$$

$\therefore \vec{e}_2$  is an eigenvector w corresponding eigenvalue 5

$$\mathbf{A}\vec{e}_3 = \begin{pmatrix} 1 & 2 & 4 \\ 0 & 1 & 0 \\ 3 & 1 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ -6 \\ 3 \end{pmatrix} = \dots = \begin{pmatrix} -6 \\ -6 \\ 3 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ -6 \\ 3 \end{pmatrix} = 1\vec{e}_3$$

$\therefore \vec{e}_3$  is an eigenvector w corresponding eigenvalue 1

**Exercise 1**

1. The matrix  $\mathbf{G} = \begin{pmatrix} -11 & 3 & -6 \\ 8 & -2 & 4 \\ r & -6 & 9 \end{pmatrix}$  has eigenvector  $\begin{pmatrix} 9 \\ -8 \\ -16 \end{pmatrix}$ . Find the value of  $r$ . (Ans:  $r = 16$ )

2. Given that  $\mathbf{A} = \begin{pmatrix} p & -3 & 0 \\ 1 & 2 & 1 \\ -1 & q & 4 \end{pmatrix}$  has eigenvectors  $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ , find the values of the corresponding eigenvalues, as well as  $p$  and  $q$ . (Ans:  $\lambda = 2, 4, 5; p = 5, q = 3$ )

3. Find the eigenvalues and corresponding eigenvectors for the matrix  $\begin{pmatrix} -3 & a & 2 \\ -2b & 2 & b \\ 0 & 2a & 1 \end{pmatrix}$ .  
(Ans:  $\lambda = -3, 1, 2; \begin{pmatrix} ab-10 \\ -4b \\ 2ab \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} a \\ 1 \\ 2a \end{pmatrix}$ )

## Exercise 1

1. The matrix  $\mathbf{G} = \begin{pmatrix} -11 & 3 & -6 \\ 8 & -2 & 4 \\ r & -6 & 9 \end{pmatrix}$  has eigenvector  $\begin{pmatrix} 9 \\ -8 \\ -16 \end{pmatrix}$ . Find the value of  $r$ . (Ans:  $r = 16$ )

Let  $\lambda$  be the eigenvalue.

$$\begin{pmatrix} -11 & 3 & -6 \\ 8 & -2 & 4 \\ r & -6 & 9 \end{pmatrix} \begin{pmatrix} 9 \\ -8 \\ -16 \end{pmatrix} = \lambda \begin{pmatrix} 9 \\ -8 \\ -16 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -99 - 24 + 96 \\ 72 + 16 - 64 \\ 9r + 48 - 144 \end{pmatrix} = \begin{pmatrix} 9\lambda \\ -8\lambda \\ -16\lambda \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -27 \\ 24 \\ 9r - 96 \end{pmatrix} = \begin{pmatrix} 9\lambda \\ -8\lambda \\ -16\lambda \end{pmatrix}$$

$$\begin{aligned} -27 &= 9\lambda \\ \Rightarrow \lambda &= -3 \end{aligned}$$

$$9r - 96 = -16\lambda$$

$$9r - 96 = -16(-3)$$

$$r = 16$$

2. Given that  $\mathbf{A} = \begin{pmatrix} p & -3 & 0 \\ 1 & 2 & 1 \\ -1 & q & 4 \end{pmatrix}$  has eigenvectors  $\begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ , find the values of the corresponding eigenvalues, as well as  $p$  and  $q$ . (Ans:  $\lambda = 2, 4, 5; p = 5, q = 3$ )

Let the corr. eigenvalues be  $\lambda_1, \lambda_2, \lambda_3$  respectively.

$$\begin{pmatrix} p & -3 & 0 \\ 1 & 2 & 1 \\ -1 & q & 4 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$\vdots$

$$\begin{pmatrix} p-3 \\ 2 \\ q-5 \end{pmatrix} = \begin{pmatrix} \lambda_1 \\ \lambda_1 \\ -\lambda_1 \end{pmatrix}$$

$$\begin{array}{l|l|l} \lambda_1 = 2 & p-3 = \lambda_1 & q-5 = -\lambda_1 \\ & p-3 = 2 & q-5 = -2 \\ & p = 5 & q = 3 \end{array}$$

$$\begin{pmatrix} 5 & -3 & 0 \\ 1 & 2 & 1 \\ -1 & 3 & 4 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$$

$\vdots$

$$\Rightarrow \begin{pmatrix} 12 \\ 4 \\ -4 \end{pmatrix} = \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$$

$$\Rightarrow 4 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix} = \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$$

$$\therefore \lambda_2 = 4$$

$$\begin{pmatrix} \phantom{5} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \lambda_3 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$\vdots$

$$5 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \lambda_3 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$\therefore \lambda_3 = 5$$

3. Find the eigenvalues and corresponding eigenvectors for the matrix  $\begin{pmatrix} -3 & a & 2 \\ -2b & 2 & b \\ 0 & 2a & 1 \end{pmatrix}$ .

$$\left( \text{Ans: } \lambda = -3, 1, 2; \begin{pmatrix} ab-10 \\ -4b \\ 2ab \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} a \\ 1 \\ 2a \end{pmatrix} \right)$$

$$\text{Let } A = \begin{pmatrix} -3 & a & 2 \\ -2b & 2 & b \\ 0 & 2a & 1 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} -3-\lambda & a & 2 \\ -2b & 2-\lambda & b \\ 0 & 2a & 1-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\Rightarrow (-3-\lambda) \begin{vmatrix} 2-\lambda & b \\ 2a & 1-\lambda \end{vmatrix} + 2b \begin{vmatrix} a & 2 \\ 2a & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (-3-\lambda) [(2-\lambda)(1-\lambda) - 2ab] + 2b(a - a\lambda - 4a) = 0$$

$$\Rightarrow (-3-\lambda) [\lambda^2 - 3\lambda + 2 - 2ab] + 2ab(-\lambda - 3) = 0$$

$$\Rightarrow (-\lambda - 3) [\lambda^2 - 3\lambda + 2 - 2ab + 2ab] = 0$$

$$\Rightarrow (-1)(\lambda + 3)(\lambda^2 - 3\lambda + 2) = 0$$

$$(\lambda + 3)(\lambda - 2)(\lambda - 1) = 0$$

$$\boxed{\lambda = -3} \text{ or } \boxed{\lambda = 2} \text{ or } \boxed{\lambda = 1}$$

For  $\lambda = -3$ :

$$A - \lambda I = \begin{pmatrix} 0 & a & 2 \\ -2b & -1 & b \\ 0 & 2a & 4 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ a \\ 2 \end{pmatrix} \times \begin{pmatrix} -2b \\ 5 \\ b \end{pmatrix} = \begin{pmatrix} ab - 10 \\ -(0 + 4b) \\ 0 + 2ab \end{pmatrix}$$

$$\therefore \text{For } \lambda = -3, \text{ an evec is } \begin{pmatrix} ab - 10 \\ -4b \\ 2ab \end{pmatrix}$$

For  $\lambda = 1$

$$A - \lambda I = \begin{pmatrix} -4 & a & 2 \\ -2b & 1 & b \\ 0 & 2a & 0 \end{pmatrix}$$

$$\begin{pmatrix} -4 \\ a \\ 2 \end{pmatrix} \times \begin{pmatrix} -2b \\ 1 \\ b \end{pmatrix} = \begin{pmatrix} ab - 2 \\ -(-4b + 4b) \\ -4 + 2ab \end{pmatrix}$$

$$= \begin{pmatrix} ab - 2 \\ 0 \\ 2ab - 4 \end{pmatrix}$$

$$= (ab - 2) \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

$$\therefore \text{For } \lambda = 1, \text{ an evec is } \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

For  $\lambda = 2$ :

$$A - \lambda I = \begin{pmatrix} -5 & a & 2 \\ -2b & 0 & b \\ 0 & 2a & -1 \end{pmatrix}$$

$$\begin{pmatrix} -2b \\ 0 \\ b \end{pmatrix} \times \begin{pmatrix} 0 \\ 2a \\ -1 \end{pmatrix} = \begin{pmatrix} 0 - 2ab \\ -(2b - 0) \\ -4ab - 0 \end{pmatrix}$$

$$= \begin{pmatrix} -2ab \\ -2b \\ -4ab \end{pmatrix}$$

$$= (-2b) \begin{pmatrix} a \\ 1 \\ 2a \end{pmatrix}$$

$$\text{For } \lambda = 2, \text{ an evec is } \begin{pmatrix} a \\ 1 \\ 2a \end{pmatrix}$$



**Matrix Algebra**

1. Let  $\mathbf{A}$  be an  $n \times n$  matrix, which has distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$  and corresponding eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ .

$$f(x) = x + k \implies f(\lambda) = \lambda + kI$$

Let  $\mathbf{B} = \mathbf{A} + k\mathbf{I}$  where  $k$  is a real constant. Then

$\mathbf{B}$  has eigenvalues  $\lambda_1 + k, \lambda_2 + k, \dots, \lambda_n + k$ , with corresponding eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ .

Notice that:

- (a)  $\mathbf{B}$  have the same eigenvectors as  $\mathbf{A}$ , and
- (b) Each eigenvalue for  $\mathbf{B}$  is  $k$  more than the eigenvalue for  $\mathbf{A}$ .

**Discussion:**

Consider the eigenvector  $\mathbf{e}_1$ . We have  $A\vec{e}_1 = \lambda_1 \vec{e}_1$  — (1)

$$\begin{aligned} \text{Matrix } B\vec{e}_1 &= (A + kI)\vec{e}_1 \\ &= A\vec{e}_1 + kI\vec{e}_1 \\ &= \lambda_1 \vec{e}_1 + k\vec{e}_1 \quad (\text{using (1)}) \\ &= (\lambda_1 + k)\vec{e}_1 \quad \text{Scalar} \end{aligned}$$

$\therefore \vec{e}_1$  is an eigenvector of  $\mathbf{B}$ , with corresponding eigenvalue  $(\lambda_1 + k)$

2. Other facts you should know: (Not related to ques above)

- (a) If matrix  $\mathbf{A}$  has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , and matrix  $\mathbf{B}$  has eigenvalue  $\mu$  and corresponding eigenvector  $\mathbf{e}$ , then  $(\mathbf{A} + \mathbf{B})\mathbf{e} = (\lambda + \mu)\mathbf{e}$ .

$$A\vec{e} = \lambda\vec{e} \quad B\vec{e} = \mu\vec{e}$$

$$\begin{aligned} \text{Matrix } (A + B)\vec{e} &= A\vec{e} + B\vec{e} \\ &= \lambda\vec{e} + \mu\vec{e} \\ &= (\lambda + \mu)\vec{e} \quad \text{Scalar (shown)} \end{aligned}$$

- (b) If matrix  $\mathbf{A}$  has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , and matrix  $\mathbf{B}$  has eigenvalue  $\mu$  and corresponding eigenvector  $\mathbf{e}$ , then  $\mathbf{A}\mathbf{B}\mathbf{e} = \lambda\mu\mathbf{e}$ . Hence, the matrix  $\mathbf{AB}$  with eigenvector  $\mathbf{e}$  has eigenvalue  $\lambda\mu$ .

$$A\vec{e} = \lambda\vec{e} \quad B\vec{e} = \mu\vec{e}$$

$$\begin{aligned} (AB)\vec{e} &= AB\vec{e} \\ &= A(\mu\vec{e}) \quad \text{Scalar} \\ &= (\lambda\vec{e})\mu \quad \text{Scalar} \\ &= \lambda\mu\vec{e} \quad \text{Scalar} \end{aligned}$$

- (c) If matrix  $A$  has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , then the matrix  $A^m$  (where  $m \in \mathbb{N}$ ) has eigenvalue  $\lambda^m$  and corresponding eigenvector  $\mathbf{e}$ .

$$A\mathbf{e} = \lambda\mathbf{e}$$

$$\begin{aligned} A^m \mathbf{e} &= (A^{m-1} A) \mathbf{e} \\ &= A^{m-1} (A \mathbf{e}) \\ &= A^{m-1} (\lambda \mathbf{e}) \\ &= \lambda (A^{m-1} \mathbf{e}) \\ &= \lambda (A^{m-2} A) \mathbf{e} \\ &= \lambda A^{m-2} (A \mathbf{e}) \\ &= \lambda A^{m-2} (\lambda \mathbf{e}) \\ &= \lambda^2 (A^{m-2} \mathbf{e}) \\ &\vdots m \text{ times} \\ &= \lambda^m (A^{m-m} \mathbf{e}) \\ &= \lambda^m (I \mathbf{e}) = \lambda^m \mathbf{e} \quad (\text{shown}) \end{aligned}$$

$$A^m \mathbf{e} = \lambda^m \mathbf{e}$$

- (d) In mathematics, a **matrix polynomial** is a polynomial with square matrices as variables.

Given an ordinary, scalar-valued polynomial

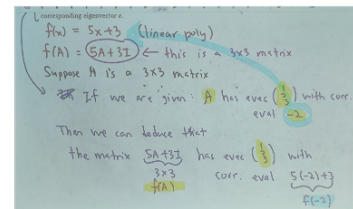
$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m$$

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_m x^m,$$

this polynomial evaluated at a matrix  $A$  is

$$f(A) = a_0 I + a_1 A + a_2 A^2 + \dots + a_m A^m,$$

where  $I$  is the **identity matrix**.



If matrix  $A$  has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , then  $f(A)$  has eigenvalue  $f(\lambda)$  and corresponding eigenvector  $\mathbf{e}$ .

$$[f(A)]\mathbf{e} = [f(\lambda)]\mathbf{e}$$

Given:  $A\mathbf{e} = \lambda\mathbf{e}$

$$\begin{aligned} [f(A)]\mathbf{e} &= [a_0 I + a_1 A + a_2 A^2 + \dots + a_m A^m] \mathbf{e} \\ &= a_0 I \mathbf{e} + a_1 A \mathbf{e} + a_2 A^2 \mathbf{e} + \dots + a_m A^m \mathbf{e} \\ &= a_0 \mathbf{e} + a_1 (\lambda \mathbf{e}) + a_2 (\lambda^2 \mathbf{e}) + \dots + a_m (\lambda^m \mathbf{e}) \\ &= (a_0 + a_1 \lambda + a_2 \lambda^2 + \dots + a_m \lambda^m) \mathbf{e} \\ &= [f(\lambda)] \mathbf{e} \end{aligned}$$

$\therefore \mathbf{e}$  is an **evect** of the matrix  $f(A)$  w/ **corr. eigenvalue**  $f(\lambda)$

### 3. Cayley-Hamilton Theorem:

$A$

If matrix  $A$  has **characteristic polynomial**  $P_A(\lambda) = \det(A - \lambda I)$ , then

$$P_A(A) = 0.$$

From this we can determine the inverse matrix of  $A$ .

Example

Suppose that

$$\det(A - \lambda I) = -\lambda^3 + 4\lambda^2 + 7\lambda + 5$$

Then by Cayley-Hamilton Theorem,

$$\begin{aligned} -A^3 + 4A^2 + 7A + 5I &= 0 \\ -A^3 + 4A^2 + 7A &= -5I \end{aligned}$$

Premultiply by  $A^{-1}$

$$\begin{aligned} -A^2 + 4A + 7I &= -5A^{-1}I \\ -A^2 + 4A + 7I &= -5A^{-1} \\ A^{-1} &= \frac{1}{-5}(-A^2 + 4A + 7I) \end{aligned}$$

Another way

Characteristic Eqn:  
 $-\lambda^3 + 4\lambda^2 + 7\lambda + 5 = 0$  (Cubic eqn.)  
 $\lambda^3 - 4\lambda^2 - 7\lambda - 5 = 0$   
 By Cayley-Hamilton, sub  $\lambda = A$   
 $A^3 - 4A^2 - 7A - 5I = 0$

**Example 14**

The matrix  $\mathbf{A} = \begin{pmatrix} -1 & 12 & 1 \\ 1 & 3 & 3 \\ 0 & 0 & -4 \end{pmatrix}$  has eigenvalues  $\lambda = -3, -4, 5$ . For each of the following cases, write down the corresponding eigenvalues and write down the relationship between each matrix and  $\mathbf{A}$ .

$$(a) \mathbf{B} = \begin{pmatrix} 1 & 12 & 1 \\ 1 & 5 & 3 \\ 0 & 0 & -2 \end{pmatrix} \quad (b) \mathbf{C} = \begin{pmatrix} -2 & 12 & 1 \\ 1 & 2 & 3 \\ 0 & 0 & -5 \end{pmatrix} \quad (c) \mathbf{D} = \begin{pmatrix} 3 & 12 & 1 \\ 1 & 7 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

**Solution**

$$a) \mathbf{B} = \begin{pmatrix} -1 & 12 & 1 \\ 1 & 3 & 3 \\ 0 & 0 & -4 \end{pmatrix} + \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \quad b) \mathbf{C} = \mathbf{A} - \mathbf{I} \quad c) \mathbf{D} = \mathbf{A} + 4\mathbf{I}$$

$$= \mathbf{A} + 2\mathbf{I}$$

$\therefore$  Eigenvalues of  $\mathbf{B}$  are

$$-3+2 = -1,$$

$$-4+2 = -2,$$

$$5+2 = 7$$

**Example 15**

The matrix  $\mathbf{A}$  is given by  $\begin{pmatrix} -6 & 2 & 3 \\ -14 & 3 & 10 \\ -4 & 2 & 1 \end{pmatrix}$ . The matrix  $\mathbf{B}$  is given by  $\mathbf{A} + 3\mathbf{I}$ .

Two eigenvalues of  $\mathbf{A}$  are  $\lambda = -2, 1$  and their corresponding eigenvectors are  $\begin{pmatrix} 5 \\ -2 \\ 8 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ .

Find the eigenvalues and corresponding eigenvectors for  $\mathbf{B}$ .

$$\left( \text{Ans: For } \mathbf{B}, \text{ eigenvalues are } 1, 4, 2; \text{ eigenvectors are } \begin{pmatrix} 5 \\ -2 \\ 8 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right)$$

**Solution**

$$(\mathbf{A} - \lambda \mathbf{I}) = \begin{pmatrix} -6-\lambda & 2 & 3 \\ -14 & 3-\lambda & 10 \\ -4 & 2 & 1-\lambda \end{pmatrix}$$

$$\text{For } \lambda = -1$$

$$\mathbf{A} - \lambda \mathbf{I} = \begin{pmatrix} -5 & 2 & 3 \\ -14 & 4 & 10 \\ -4 & 2 & 2 \end{pmatrix}$$

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\vdots$$

$$\lambda^3 + 2\lambda^2 - \lambda - 2 = 0$$

$$(\lambda + 2)(\lambda - 1)(\lambda + 1) = 0$$

$$\therefore \lambda = -2 \text{ or } \lambda = 1 \text{ or } \lambda = -1$$

$$\begin{pmatrix} -5 \\ 2 \\ 3 \end{pmatrix} \times \begin{pmatrix} -4 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 4-6 \\ -(-10+12) \\ -10+8 \end{pmatrix} = \begin{pmatrix} -2 \\ -2 \\ -2 \end{pmatrix} = (-2) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\therefore \text{For } \lambda = -1, \text{ an evvec is } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

## Solution for Example 15 (cont.)

$\therefore$  Evec for B is

$$\begin{pmatrix} 5 \\ -2 \\ 8 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$\therefore$  Since  $B = A + 3I$

eval of B is

$$\lambda_1 = -2 + 3 = 1 \neq$$

$$\lambda_2 = 1 + 3 = 4 \neq$$

$$\lambda_3 = -1 + 3 = 2 \neq$$

**Example 16**

The matrix  $A$  is given by  $\begin{pmatrix} -3 & 0 \\ 5 & 2 \end{pmatrix}$ . Determine the eigenvalues and corresponding eigenvectors of the matrix  $A^6$ .

$$\left( \text{Ans: } \lambda = 729, \begin{pmatrix} 1 \\ -1 \end{pmatrix}; \lambda = 64, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right)$$

**Solution**

$$A = \begin{pmatrix} -3 & 0 \\ 5 & 2 \end{pmatrix}$$

$$A - \lambda I = \begin{pmatrix} -3-\lambda & 0 \\ 5 & 2-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$(-3-\lambda)(2-\lambda) - 0 = 0$$

$$-6 - 2\lambda + 3\lambda + \lambda^2 = 0$$

$$\lambda^2 + \lambda - 6 = 0$$

$$(\lambda + 3)(\lambda - 2) = 0$$

$$\lambda = -3 \text{ or } \lambda = 2$$

$$(A - \lambda I)\vec{x} = \vec{0}$$

$$\Rightarrow \begin{pmatrix} -3-\lambda & 0 \\ 5 & 2-\lambda \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{cases} (-3-\lambda)x = 0 \\ 5x + (2-\lambda)y = 0 \end{cases}$$

For  $\lambda = -3$

$$\begin{cases} 0 = 0 \\ 5x + 5y = 0 \end{cases} \quad \therefore \text{For } \lambda = -3, \text{ an evec is } \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

$$\begin{aligned} x + y &= 0 \\ x &= -y \end{aligned}$$

set "free"

For  $\lambda = 2$ :

$$\begin{cases} -5x = 0 \\ 5x = 0 \end{cases} \quad \therefore \text{For } \lambda = 2, \text{ an evec is } \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

We have  $x = 0$

Since there is no restriction on  $y$ ,  $y$  is the free variable

For  $A^6$

The eval are

$$(-3)^6 = 729 \text{ and } 2^6 = 64$$

The corr. evec are  $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$  and  $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$  respectively

**Example 17**

Given that  $A = \begin{pmatrix} -7 & k & -8 \\ 2 & l & m \\ 5 & 4 & 6 \end{pmatrix}$ , and that two eigenvectors are  $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ , find the values of  $k, l, m$ , the eigenvalues and the last eigenvector.

$$\left( \text{Ans: } k = -4, l = -1, m = 2; \lambda_1 = -2, \lambda_2 = -1, \lambda_3 = 1; \text{last eigenvector is } \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right)$$

Hence, determine the eigenvalues and eigenvectors of  $B = A + 2A^3$ .

$$\left( \text{Ans: eigenvalues are } -18, -3, 3; \text{eigenvectors are } \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right)$$

**Solution**

For the eigenvectors  $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$ , let the eigenvalues be  $\lambda_1$  and  $\lambda_2$  respectively

$$A \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} = \lambda_1 \begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix} \quad ; \quad A \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} = \lambda_2 \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$$

Solve simul eqns  
to find  $k, l, m, \lambda_1, \lambda_2$ .

$$A - \lambda I = \begin{pmatrix} -7-\lambda & -4 & -8 \\ 2 & -1-\lambda & 2 \\ 5 & 4 & 6-\lambda \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\lambda^3 + 2\lambda^2 - \lambda - 2 = 0$$

$\lambda = -2$  and  $\lambda = -1$  are  
2 soln of this cubic eqn

Find the 3rd  $\lambda$  value  
using a valid method

$$\lambda_3 = 1$$

For  $\lambda = 1$ ,

$$A - \lambda I = \begin{pmatrix} -8 & -4 & -8 \\ 2 & -2 & 2 \\ 5 & 4 & 5 \end{pmatrix}$$

Take cross product of any 2 non-parallel rows

$$\text{For } \lambda = 1, \text{ an evect is } \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$B = A + 2A^3 \Rightarrow f(A), \text{ where } f(x) = x + 2x^3$$

For  $B$ , the eval are

$$-2 + 2(-2)^3 = -18,$$

$$-1 + 2(-1)^3 = -3,$$

$$1 + 2(1)^3 = 3$$

$\therefore$  The corr. evect are  $\begin{pmatrix} 0 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$  respectively

**Solution for Example 17** (cont.)

**Example 18***Triangular matrix*

The matrix  $\mathbf{A}$  is given as  $\begin{pmatrix} 1 & 2 & -1 \\ 0 & 2 & 4 \\ 0 & 0 & -3 \end{pmatrix}$ . Determine the characteristic equation of  $\mathbf{A}$  and use it to determine the matrix  $\mathbf{A}^{-1}$ .

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\left( \text{Ans: } \mathbf{A}^3 - 7\mathbf{A} + 6\mathbf{I} = \mathbf{0}, \mathbf{A}^{-1} = \begin{pmatrix} 1 & -1 & -\frac{5}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} \\ 0 & 0 & -\frac{1}{3} \end{pmatrix} \right)$$

**Solution**

The characteristic eqn of  $\mathbf{A}$ :

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\begin{vmatrix} 1-\lambda & 2 & -1 \\ 0 & 2-\lambda & 4 \\ 0 & 0 & -3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(2-\lambda)(-3-\lambda) = 0$$

$$\vdots$$

$$\boxed{\lambda^3 - 7\lambda + 6 = 0}$$

By Cayley-Hamilton Thm,

$$\mathbf{A}^3 - 7\mathbf{A} + 6\mathbf{I} = \mathbf{0}$$

$$\Rightarrow \mathbf{A}^3 - 7\mathbf{A} = -6\mathbf{I}$$

$$\Rightarrow \mathbf{A}^{-1}\mathbf{A}^3 - 7\mathbf{A}^{-1}\mathbf{A} = -6\mathbf{A}^{-1}\mathbf{I}$$

$$\Rightarrow \mathbf{A}^2 - 7\mathbf{I} = -6\mathbf{A}^{-1}$$

$$\Rightarrow \boxed{\mathbf{A}^{-1} = -\frac{1}{6}(\mathbf{A}^2 - 7\mathbf{I})}$$

$$\vdots$$

$$= \begin{pmatrix} 1 & -1 & -\frac{5}{3} \\ 0 & \frac{1}{2} & \frac{2}{3} \\ 0 & 0 & -\frac{1}{3} \end{pmatrix}$$



*Exercise 2*

1. The matrix  $\mathbf{A}$  has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , and the matrix  $\mathbf{B}$  has eigenvalue  $\mu$  and corresponding eigenvector  $\mathbf{e}$ . Show that the matrices  $\mathbf{AB}$  and  $\mathbf{BA}$  have the same eigenvalues and corresponding eigenvectors.

2. The matrix  $\mathbf{A} = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 5 & 0 \\ 0 & 3 & -6 \end{pmatrix}$  has eigenvectors  $\begin{pmatrix} 1 \\ 0 \\ -10 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 11 \\ 3 \end{pmatrix}$

(a) Find the eigenvalues of  $\mathbf{A}^3$ .

(Ans: 64, 125, -216)

(b) Given that the matrix  $\mathbf{B} = \mathbf{A} - \mathbf{A}^2$ . Find the eigenvalues and eigenvectors of  $\mathbf{B}$ .

(Ans:  $\lambda = -12, -20, -42; \begin{pmatrix} 1 \\ 0 \\ -10 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 11 \\ 3 \end{pmatrix}$ )

3. If a matrix,  $\mathbf{A}$ , has eigenvalue  $\lambda$  and corresponding eigenvector  $\mathbf{e}$ , show the following:

(a)  $\mathbf{A}\mathbf{e} + \mathbf{A}^2\mathbf{e} = (\lambda + \lambda^2)\mathbf{e}$

(b)  $\mathbf{A}\mathbf{e} + \mathbf{A}^{-1}\mathbf{e} = \left(\lambda + \frac{1}{\lambda}\right)\mathbf{e}$

4. Using the Cayley-Hamilton theorem, find the inverse of the matrix  $\mathbf{A} = \begin{pmatrix} 0 & -4 & -6 \\ -1 & 0 & -3 \\ 1 & 2 & 5 \end{pmatrix}$ .

(Ans:  $\mathbf{A}^{-1} = \begin{pmatrix} \frac{3}{2} & 2 & 3 \\ \frac{1}{2} & \frac{3}{2} & \frac{3}{2} \\ -\frac{1}{2} & -1 & -1 \end{pmatrix}$ )

## Exercise 2

1. The matrix **A** has eigenvalue  $\lambda$  and corresponding eigenvector **e**, and the matrix **B** has eigenvalue  $\mu$  and corresponding eigenvector **e**. Show that the matrices **AB** and **BA** have the same eigenvalues and corresponding eigenvectors.

$$\text{Given } A\vec{e} = \lambda\vec{e} ; B\vec{e} = \mu\vec{e}$$

$$\begin{aligned} AB\vec{e} &= A(B\vec{e}) & BA\vec{e} &= B(A\vec{e}) \\ &= A(\mu\vec{e}) & &= B(\lambda\vec{e}) \\ &= \mu(A\vec{e}) & &= \lambda(B\vec{e}) \\ &= (\mu\lambda)\vec{e} & &= \lambda\mu\vec{e} \\ & & &= (\mu\lambda)\vec{e} \end{aligned}$$

$\therefore$  AB and BA have the same eigenvalue of  $\mu\lambda$  and corr. vec of  $\vec{e}$

3. If a matrix, **A**, has eigenvalue  $\lambda$  and corresponding eigenvector **e**, show the following:

(a)  $A\mathbf{e} + A^2\mathbf{e} = (\lambda + \lambda^2)\mathbf{e}$

(b)  $A\mathbf{e} + A^{-1}\mathbf{e} = \left(\lambda + \frac{1}{\lambda}\right)\mathbf{e}$

a)  $\text{Given : } A\vec{e} = \lambda\vec{e}$

Premultiply both sides by A

$$\begin{aligned} AA\vec{e} &= A\lambda\vec{e} \\ \Rightarrow A^2\vec{e} &= \lambda(A\vec{e}) \\ \Rightarrow A^2\vec{e} &= \lambda(\lambda\vec{e}) \\ \Rightarrow A^2\vec{e} &= \lambda^2\vec{e} \end{aligned}$$

$$\begin{aligned} \therefore A\vec{e} + A^2\vec{e} &= \lambda\vec{e} + \lambda^2\vec{e} \\ &= (\lambda + \lambda^2)\vec{e} \end{aligned}$$

b)  $\text{Given : } A\vec{e} = \lambda\vec{e}$

Premultiply both sides by  $A^{-1}$

$$\begin{aligned} A^{-1}A\vec{e} &= A^{-1}\lambda\vec{e} \\ \lambda(A^{-1}\vec{e}) &= \vec{e} \\ A^{-1}\vec{e} &= \frac{1}{\lambda}\vec{e} \end{aligned}$$

$$\begin{aligned} \therefore A\vec{e} + A^{-1}\vec{e} &= \lambda\vec{e} + \frac{1}{\lambda}\vec{e} \\ &= \left(\lambda + \frac{1}{\lambda}\right)\vec{e} \end{aligned}$$

Diagonalisation1. **Definition (Diagonalisable matrix):**

A square matrix  $\mathbf{A}$  is said to be **diagonalisable** if  $\mathbf{A} = \mathbf{PDP}^{-1}$  for some **invertible** matrix  $\mathbf{P}$  and some diagonal matrix  $\mathbf{D}$ .

2. **Fact:** All  $n \times n$  matrices which have  $n$  **distinct eigenvalues** are **diagonalisable**.3. **Fact:**

Let  $\mathbf{A}$  be an  $n \times n$  matrix with  $n$  distinct eigenvalues  $\lambda_1, \lambda_2, \dots, \lambda_n$  and corresponding eigenvectors  $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ .

Let

$$\mathbf{P} = (\mathbf{e}_1 \quad \mathbf{e}_2 \quad \mathbf{e}_3 \quad \dots \quad \mathbf{e}_n),$$

i.e., the **columns** of  $\mathbf{P}$  are the  $n$  **eigenvectors** of  $\mathbf{A}$ , and

$$\mathbf{P} = \begin{pmatrix} \boxed{\phantom{0}} & \boxed{\phantom{0}} & \boxed{\phantom{0}} \\ \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{pmatrix}$$

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix},$$

$$\begin{pmatrix} \boxed{1} & \boxed{4} & \boxed{7} \\ \boxed{2} & \boxed{5} & \boxed{8} \\ \boxed{3} & \boxed{6} & \boxed{9} \end{pmatrix} \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} = \begin{pmatrix} 1\lambda_1 & 4\lambda_2 & 7\lambda_3 \\ 2\lambda_1 & 5\lambda_2 & 8\lambda_3 \\ 3\lambda_1 & 6\lambda_2 & 9\lambda_3 \end{pmatrix}$$

$\mathbf{e}_1\lambda_1 \quad \mathbf{e}_2\lambda_2 \quad \mathbf{e}_3\lambda_3$

i.e.,  $\mathbf{D}$  is a diagonal matrix whose **diagonal entries** are **eigenvalues** of  $\mathbf{A}$  that correspond, respectively, to the eigenvectors in  $\mathbf{P}$ .

$$\mathbf{AP} = \begin{cases} \lambda_1 \mathbf{e}_1 \\ \lambda_2 \mathbf{e}_2 \\ \lambda_3 \mathbf{e}_3 \end{cases} \therefore \mathbf{AP} = \mathbf{PD}$$

Then it *must* be true that  $\mathbf{P}$  is invertible, and that  $\mathbf{AP} = \mathbf{PD}$ . Hence,  $\mathbf{A} = \mathbf{PDP}^{-1}$ .

## 4. Note that

$$\mathbf{D}^m = \begin{pmatrix} \lambda_1 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & 0 & \dots & 0 \\ 0 & 0 & \lambda_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n \end{pmatrix}^m = \begin{pmatrix} \lambda_1^m & 0 & 0 & \dots & 0 \\ 0 & \lambda_2^m & 0 & \dots & 0 \\ 0 & 0 & \lambda_3^m & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \lambda_n^m \end{pmatrix}$$

so it is easy to find powers of  $\mathbf{D}$ .

5. **Finding powers of square matrices:**

Consider a diagonalisable square matrix  $\mathbf{A}$  that can be written in the form  $\mathbf{A} = \mathbf{PDP}^{-1}$ .

Suppose that we wish to determine  $\mathbf{A}^{20}$ .

Then

$$\begin{aligned} \mathbf{A}^{20} &= (\mathbf{PDP}^{-1})^{20} \\ &= (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \dots (\mathbf{PDP}^{-1})(\mathbf{PDP}^{-1}) \\ &= \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P})\mathbf{D}(\mathbf{P}^{-1}\mathbf{P}) \dots \mathbf{PD}(\mathbf{P}^{-1}\mathbf{P})\mathbf{DP}^{-1} \\ &= \mathbf{PD}^{20}\mathbf{P}^{-1} \end{aligned}$$

Note that  $\mathbf{D}^{20}$  is easy to find.

Additional fact

If  $\mathbf{A}$  can be written as  $\mathbf{PDP}^{-1}$ , i.e.  $\mathbf{A} = \mathbf{PDP}^{-1}$

then,  $\mathbf{A}^{-1} = [\mathbf{PDP}^{-1}]^{-1}$

$$= (\mathbf{P}^{-1})^{-1} \mathbf{D}^{-1} \mathbf{P}^{-1}$$

$$= \mathbf{PD}^{-1}\mathbf{P}^{-1}$$

$$\boxed{\mathbf{A}^{-1} = \mathbf{PD}^{-1}\mathbf{P}^{-1}}$$

$$\mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -5 \end{pmatrix}$$

$$\mathbf{D}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \\ 0 & 0 & -1/5 \end{pmatrix}$$

Recall  
 $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$

**Example 19**

A matrix is given as  $\mathbf{B} = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ . It is known to have eigenvalues  $\lambda = 1, 2, 3$  and corresponding eigenvectors

$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ . Find  $\mathbf{B}^6$ .

$$\left( \text{Ans: } \mathbf{B}^6 = \begin{pmatrix} 1 & 728 & 602 \\ 0 & 729 & 665 \\ 0 & 0 & 64 \end{pmatrix} \right)$$

**Solution**

$$\text{Let } \mathbf{D} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \mathbf{P} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 0 \end{pmatrix}$$

$$\mathbf{D}^6 = \begin{pmatrix} 1^6 & 0 & 0 \\ 0 & 2^6 & 0 \\ 0 & 0 & 3^6 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 729 \end{pmatrix}$$

Find  $\mathbf{P}^{-1}$

$\vdots$

$$\mathbf{P}^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ 0 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\mathbf{B}^6 = \mathbf{P} \mathbf{D}^6 \mathbf{P}^{-1}$$

$\vdots$

Ans  $\neq$

**Example 20**

Given that one of the eigenvectors of the matrix  $A = \begin{pmatrix} 2 & -5 & 0 \\ 1 & a & 3 \\ 0 & 0 & 5 \end{pmatrix}$  is  $\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$ , find matrices  $P$  and  $E$  such that  $A^5 = PEP^{-1}$ . (You are *not* required to find  $P^{-1}$ .)

$$\boxed{E = D^5}$$

$$\left( \text{Ans: } P = \begin{pmatrix} 1 & 5 & 15 \\ 1 & 1 & -9 \\ 0 & 0 & -32 \end{pmatrix}, E = \begin{pmatrix} -243 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3125 \end{pmatrix} \right)$$

Solution

Let  $\lambda_1$  be the eval corr. to eval  $\begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$

$$A \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix}$$

$\vdots$

$$\lambda_1 = 1, a = -4$$

$$A - \lambda I = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$\det(A - \lambda I) = 0$$

$\vdots$

$$\lambda = 1, 5, -3$$

For  $\lambda = 5$

$$A - \lambda I = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

$$\begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} \times \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix} = \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

For  $\lambda = -3$

$\vdots$

$$\therefore P = \begin{pmatrix} 5 & \boxed{\phantom{00}} & \boxed{\phantom{00}} \\ 1 & \boxed{\phantom{00}} & \boxed{\phantom{00}} \\ 0 & \boxed{\phantom{00}} & \boxed{\phantom{00}} \end{pmatrix}, D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix}$$

$$E = D^5 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 5^5 & 0 \\ 0 & 0 & (-3)^5 \end{pmatrix}$$

$\vdots$

$$= \begin{pmatrix} & & \\ & & \\ & & \end{pmatrix}$$

\*\*\*\*\* END OF SECTION I \*\*\*\*\*